

D886

Secondary Science Teaching Methods

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A teacher in a school with block scheduling gives students a question about how the angle of a ramp affects a ball's speed. The class uses carts bought with a grant. Students design their own procedure and decide how to record data. Which level of inquiry is this?

- A. Confirmation inquiry
- B. Structured inquiry
- C. Open inquiry
- D. Guided inquiry

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A lab handout lists the question, the step-by-step procedure, and the expected result that students should confirm. Which inquiry level is this?

- A. Open inquiry
- B. Guided inquiry
- C. Confirmation inquiry
- D. Not inquiry at all, because any given procedure means it cannot be inquiry.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Midway through a unit, a teacher gives a five-question quiz on the digital platform the district adopted this year. The teacher then regroups students and reteaches the concept most missed. The quiz takes about 10 minutes. What type of assessment is this?

- A. Formative
- B. Summative
- C. Diagnostic of prior knowledge before any teaching
- D. Summative, because it is a quiz with a score.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

At the end of a unit, a teacher gives a lab practical and records it as a final grade for the unit. What type of assessment is this?

- A.** Summative
- B.** Formative
- C.** Formative, because it is hands-on.
- D.** Neither, because lab practicals are never assessments.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Many students in a class that meets right before lunch say heavier objects always fall faster. The class has a new set of stopwatches. Which teacher move best addresses this misconception?

- A.** Avoid the topic so students are not confused.
- B.** Have students time falls with the new stopwatches without making predictions.
- C.** Have students predict, then drop a heavy and light ball together and discuss why they land at the same time.
- D.** Tell students the correct answer once and move on.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

During a debate club practice held in the science room, a student says evolution is just a theory, so it has not been proven like the law of gravity. The club meets twice a month. Which response best reflects the nature of science?

- A.** Laws are always more important than theories.
- B.** Right; once evolution has enough proof it will become a law.
- C.** Right; theories are educated guesses.
- D.** A theory is a well-tested explanation of why something happens; a law describes what happens. Theories do not become laws.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Newton's law of universal gravitation gives the force between two masses. What does it not provide?

- A.** A description of the pattern
- B.** A way to calculate the force
- C.** An explanation of why masses attract
- D.** Anything, because laws are unproven theories.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher is teaching students to write a scientific explanation for the first time. The class includes several students who play in the school band. Which sequence best fits gradual release?

- A.** Keep giving the full template all year.
- B.** Model a sample explanation, write one together, then have students write their own.
- C.** Model an example and then give a graded test.
- D.** Have students write on their own first, then model an example.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Students measure how the temperature of water (10, 20, 30, 40, 50 degrees C) affects how fast a sugar cube dissolves. They record data on tablets donated by a local business. The business also sponsors the science fair. Which graph fits the data best?

- A.** A line graph with temperature on the x-axis and dissolving time on the y-axis
- B.** A bar graph with one bar for each temperature
- C.** A pie chart of dissolving times
- D.** A line graph with dissolving time on the x-axis

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Students compare the average height of plants grown in three soil types: sand, clay, and potting mix. Which graph fits?

- A.** A scatter plot, because every data set needs one
- B.** A bar graph with soil type on the x-axis
- C.** A bar graph with plant height on the x-axis
- D.** A line graph connecting the three soil types

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a chemistry lab group of four, one student does every step while the others watch. The group sits at the table closest to the window. Which change best improves the group's learning?

- A.** Move the group to a different table.
- B.** Let the strongest student keep doing the lab so it is done right.
- C.** Grade only the group's final lab report.
- D.** Assign rotating roles (such as materials manager, recorder, and checker) and have each student explain the results.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A teacher wants students to study radioactive decay and half-life. The school has a limited budget and a lab that was updated five years ago. Which approach fits best?

- A.** Use real radioactive samples so the lab is authentic.
- B.** Use a simulation in which students set sample sizes, predict, and graph decay over many half-lives.
- C.** Skip the topic, because virtual labs never teach real science.
- D.** Show a video of a scientist doing the lab with no student analysis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A teacher ~~in a school with block scheduling~~ gives students a question about how the angle of a ramp affects a ball's speed. ~~The class uses carts bought with a grant.~~ Students design their own procedure and decide how to record data. Which level of inquiry is this?

Target word: inquiry level **Decides:** The teacher gives only the question.

A	Inverted operation	Confirmation gives question, method, and expected answer.
B	Confusion set	In structured inquiry the teacher also gives the procedure.
C	Adjacent category	In open inquiry students also pose the question.
D	Correct	The teacher gives the question; students design the method.

2. Answer C Concept 1 • transfer

A lab handout lists the question, the step-by-step procedure, and the expected result that students should confirm. Which inquiry level is this?

Target word: inquiry level **Decides:** All three pieces are given.

A	Inverted operation	Open inquiry gives students the most control, not the least.
B	Confusion set	Guided gives only the question.
C	Correct	Students verify a known result.
D	Absolute language	Confirmation is a recognized, if low, level of inquiry.

3. Answer A Concept 2

Midway through a unit, a teacher gives a five-question quiz ~~on the digital platform the district adopted this year.~~ The teacher then regroups students and reteaches the concept most missed. ~~The quiz takes about 10 minutes.~~ What type of assessment is this?

Target word: assessment purpose **Decides:** Results are used to adjust teaching during the unit.

A	Correct	It informs instruction while learning is happening.
B	Confusion set	Summative judges learning at the end.
C	Adjacent category	A pre-assessment happens before instruction; this is midway.
D	Dropped element	Drops the purpose; format alone does not decide it.

4. Answer A Concept 2 • transfer

At the end of a unit, a teacher gives a lab practical and records it as a final grade for the unit. What type of assessment is this?

Target word: assessment purpose **Decides:** It judges learning at the end.

A	Correct	It measures learning at the end for a grade.
B	Inverted operation	Formative is used to adjust teaching during learning.
C	Dropped element	Format does not decide type; purpose and timing do.
D	Absolute language	Lab practicals can be assessments.

5. Answer C Concept 3

Many students in a class ~~that meets right before lunch~~ say heavier objects always fall faster. ~~The class has a new set of stopwatches.~~ Which teacher move best addresses this misconception?

Target word: best move **Decides:** Evidence that conflicts with the idea.

A	Inverted operation	Leaves the misconception in place.
B	Dropped element	Drops the prediction that brings out the idea.
C	Correct	A discrepant event makes students confront their idea.
D	Confusion set	Telling alone rarely changes a misconception.

6. Answer D Concept 4

During a debate club practice ~~held in the science room~~, a student says evolution is just a theory, so it has not been proven like the law of gravity. ~~The club meets twice a month.~~ Which response best reflects the nature of science?

Target word: theory vs. law **Decides:** Theories explain; laws describe; one does not become the other.

A	Absolute language	Neither outranks the other; they do different jobs.
B	Confusion set	Theories do not graduate into laws.
C	Adjacent category	That describes a hypothesis, not a theory.
D	Correct	They are different kinds of knowledge.

7. Answer C Concept 4 • transfer

Newton's law of universal gravitation gives the force between two masses. What does it not provide?

Target word: what a law lacks **Decides:** Laws describe; theories explain.

A	Inverted operation	Describing the pattern is what laws do.
B	Inverted operation	That is exactly what the law provides.
C	Correct	A law describes the pattern; explaining why is the job of a theory.
D	Confusion set	Laws are not a lower form of theory.

8. Answer B Concept 5

A teacher is teaching students to write a scientific explanation for the first time. ~~The class includes several students who play in the school band.~~ Which sequence best fits gradual release?

Target word: order of scaffolding **Decides:** Model, then guided practice, then independent.

A	Absolute language	Scaffolds should fade as skill grows.
B	Correct	I do, we do, you do.
C	Dropped element	Drops the guided practice step.
D	Inverted operation	Reverses the sequence.

9. Answer A Concept 6

Students measure how the temperature of water (10, 20, 30, 40, 50 degrees C) affects how fast a sugar cube dissolves. They record data ~~on tablets donated by a local business. The business also sponsors the science fair.~~ Which graph fits the data best?

Target word: graph type **Decides:** Temperature is a continuous independent variable.

A	Correct	Continuous independent variable on the x-axis.
B	Confusion set	Bar graphs fit categories; temperature is continuous.
C	Adjacent category	Pie charts show parts of a whole, not a relationship.
D	Inverted operation	Puts the dependent variable on the x-axis.

10. Answer B Concept 6 · transfer

Students compare the average height of plants grown in three soil types: sand, clay, and potting mix. Which graph fits?

Target word: graph for categories **Decides:** Soil type is categorical.

A	Absolute language	Scatter plots fit two numeric variables.
B	Correct	Categories are shown as separate bars.
C	Inverted operation	The independent variable belongs on the x-axis.
D	Confusion set	A line implies values between categories, which do not exist.

11. Answer D Concept 7

In a chemistry lab group of four, one student does every step while the others watch. ~~The group sits at the table closest to the window.~~ Which change best improves the group's learning?

Target word: structure for cooperation **Decides:** Roles plus individual accountability.

A	Camouflage	Seating location does not change who does the work.
B	Inverted operation	Concentrates the work instead of sharing it.
C	Dropped element	A single group grade drops individual accountability.
D	Correct	Roles and individual accountability spread the thinking.

12. Answer B Concept 8

A teacher wants students to study radioactive decay and half-life. ~~The school has a limited budget and a lab that was updated five years ago.~~ Which approach fits best?

Target word: virtual vs. physical **Decides:** Radioactive materials are hazardous.

A	Inverted operation	Radioactive materials are unsafe for a school lab.
B	Correct	A simulation handles hazards and long time spans while keeping students analyzing.
C	Absolute language	Well-designed simulations support real practices.
D	Dropped element	Drops the planning, predicting, and analyzing.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Levels of inquiry	Inquiry ranges from confirmation (question, method, and answer given) to structured (question and method given), guided (only the question given), and open (students pose the question).	Calling any hands-on lab inquiry, or mixing up guided and structured.	Who supplies the question, the procedure, and the expected answer.
2	Formative vs. summative assessment	Formative assessment happens during learning to adjust teaching; summative assessment judges learning at the end.	Classifying by format (quiz, lab) rather than by purpose and timing.	Whether results are used to change the next lesson or to give a final grade.
3	Surfacing and confronting misconceptions	Students hold preconceptions; teaching works best when it draws out those ideas and gives evidence that conflicts with them, such as a discrepant event.	Thinking that stating the correct fact once will replace a misconception.	A common wrong idea stated by students and a choice of teacher moves.
4	Theories vs. laws	A law describes what happens (often as a pattern or equation); a theory is a well-tested explanation of why; theories do not turn into laws with more proof.	Thinking a theory is a guess that becomes a law once proven.	A student statement such as it is just a theory.
5	Gradual release and scaffolding	Scaffolds such as modeling and guided practice are given first and then removed so students can perform a practice on their own.	Keeping every scaffold forever, or removing all support on day one.	A plan that moves from I do to we do to you do.
6	Choosing a graph for the data	Use a line graph (or scatter plot) when the independent variable is continuous and a bar graph when it is categorical; the independent variable goes on the x-axis.	Using a line graph for categories, or putting the dependent variable on the x-axis.	A data table with the type of independent variable named.
7	Structuring cooperative lab groups	Group work supports learning when each student has a defined role and is individually accountable, and the task requires everyone's contribution.	Assuming putting students at the same table makes them cooperate.	A lab group where one student does all the work.
8	When to use virtual labs	Simulations are the right choice when a real investigation is too dangerous, costly, slow, or at an unobservable scale; they should still require students to plan, predict, and analyze.	Thinking virtual labs are always weaker, or using them to replace every safe, doable hands-on lab.	A lab goal that involves radioactivity, very long time spans, or molecules.

Course scope (public description): This course focuses on equipping secondary science educators with the essential knowledge and skills to effectively teach science through a three-dimensional approach, integrating science and engineering practices, crosscutting concepts, and disciplinary core ideas. Participants will explore general considerations for science instruction, including inquiry-based learning, hands-on activities, and assessment strategies. By building on foundational knowledge in Three-Dimensional Science and Engineering, educators will enhance their ability to engage students in meaningful and authentic scientific learning experiences. This methods course will count towards 7 hours of the 51 total clinical hours that learners will gain from their SCED methods courses.